Communications & Networks

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Duplex, Multi-Access, Multiplex

Xiaofeng Zhong

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Reference

- Larry L. Peterson, Bruce S. Davie, Computer Networks: A Systems Approach, 3rd ed. Morgan Kaufmann Publishers, 2003.
  - Chapter 1.1, 1.2, 1.3, 1.5, 2.1, 2.7, 2.8, 3.1, 3.2, 4.1, 4.2
- Anurag Kumar, D. Manjunath, Joy Kuri, COMMUNICATION NETWORKING: *An Analytical Approach,* Morgan Kaufmann Publishers
  - Chapter 2.1~2.5

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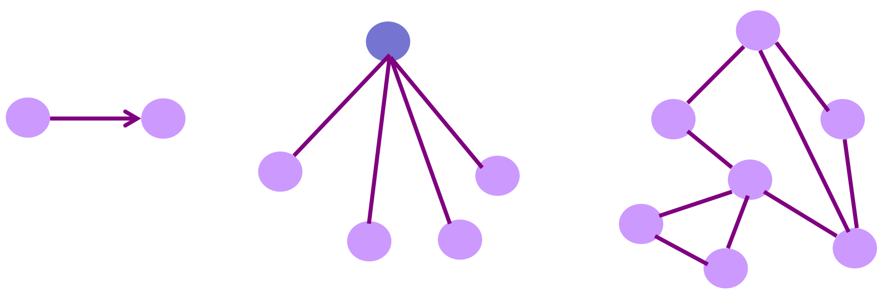
Communications to *Networks*

Scenarios: point 2 point, point 2 points, points 2 points



Transmitting

Media Access
Duplex
Multiplex
Multi-Access

Network
Switch
Routing
Transport

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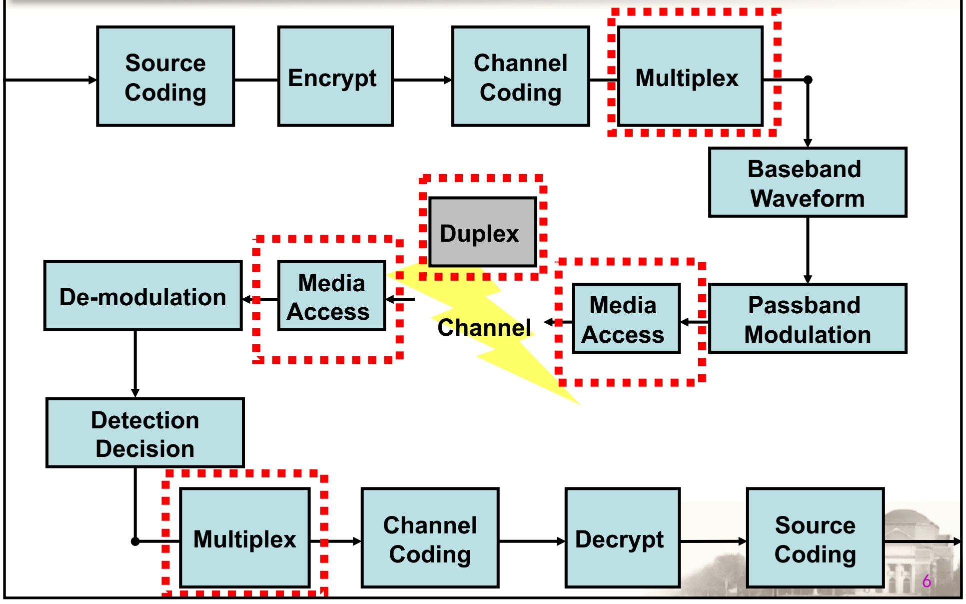
System View



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Contents

- Concepts: Duplex, Multiplex, Media Access
- Resource Divide Modes
- Multi-Access: Fix-allocation, Controlled Access, Random Access
- Multiplex: Time-multiplex (SDH)

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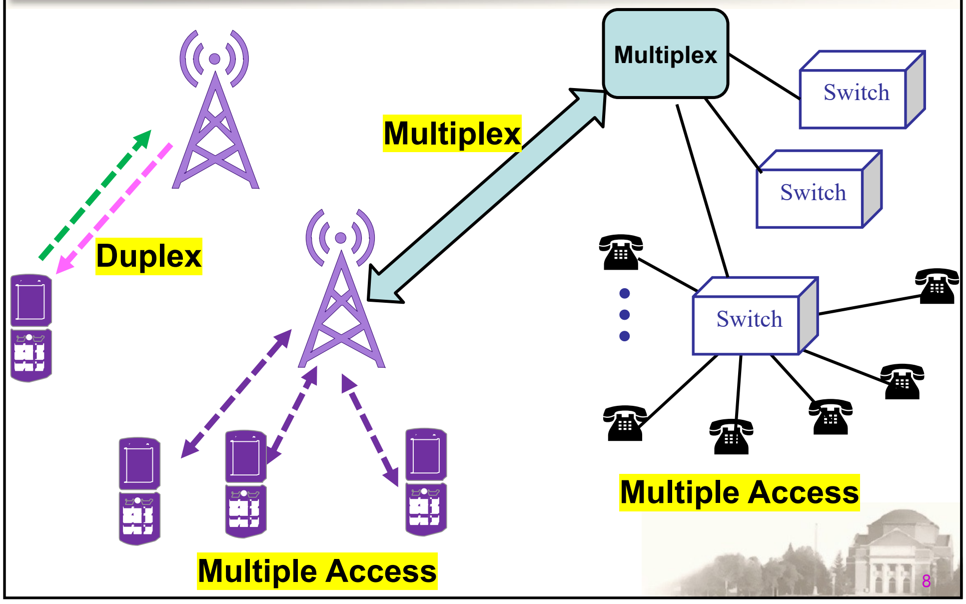
3 Concepts



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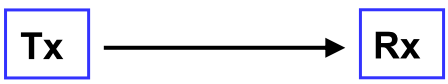
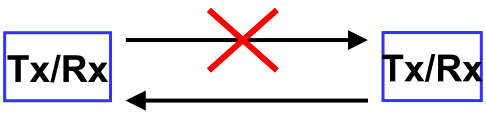
Duplex

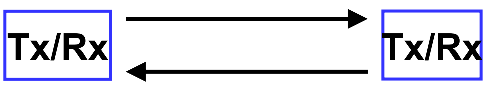
- Bi-direction Tx/Rx
- Share the channel



- Half-duplex (walkie-talkie, push-to-talk)



- Full-duplex (cellular phone, phone)



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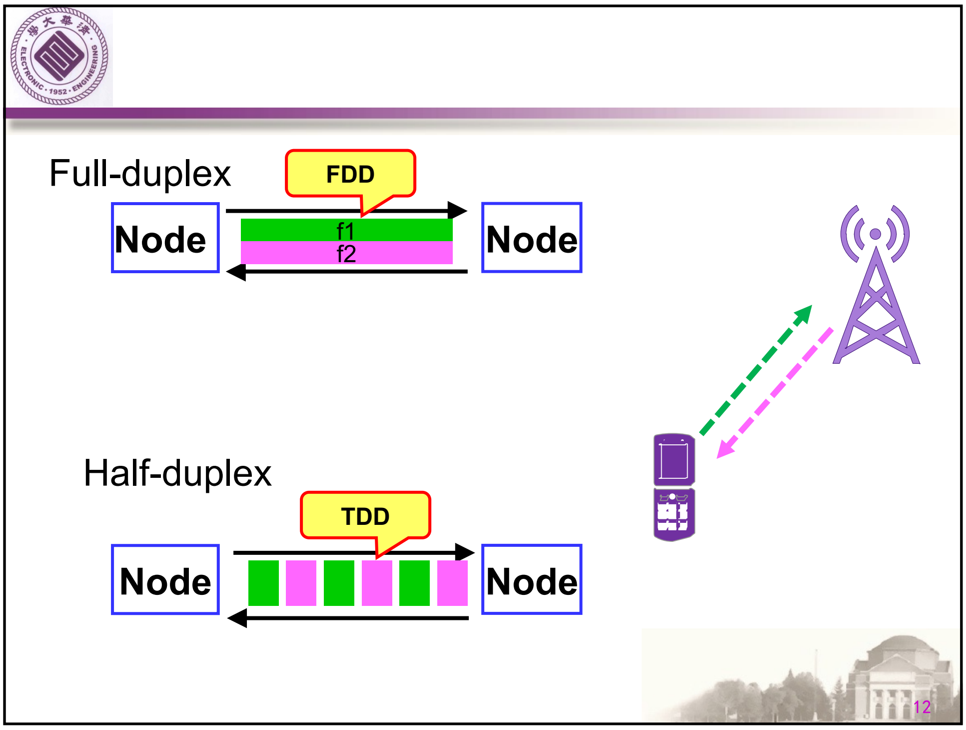

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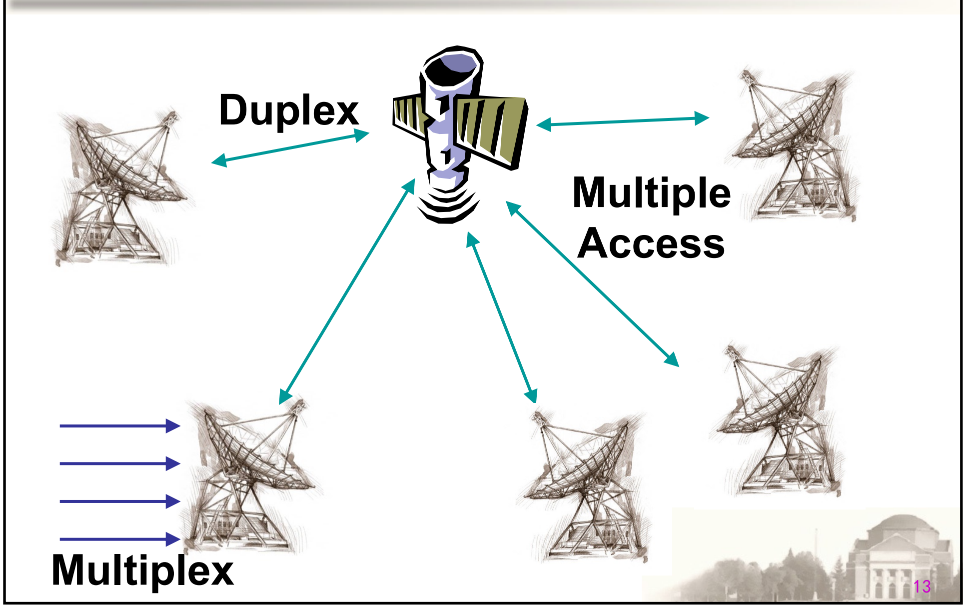
Multi-Access & Multiplex



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Multi-Access vs. Multiplex

Multiplexing

- Multiple low bit rate links Merge to one high bit rate link
- Enough Traffic and almost keep stable (about full load or even over load)
- On backbone/core network, or inter-networks

Multi-Access

- Multiple nodes/users
- Share the media (transmitting resources)
- Random arrive traffic
- On network edge, access network

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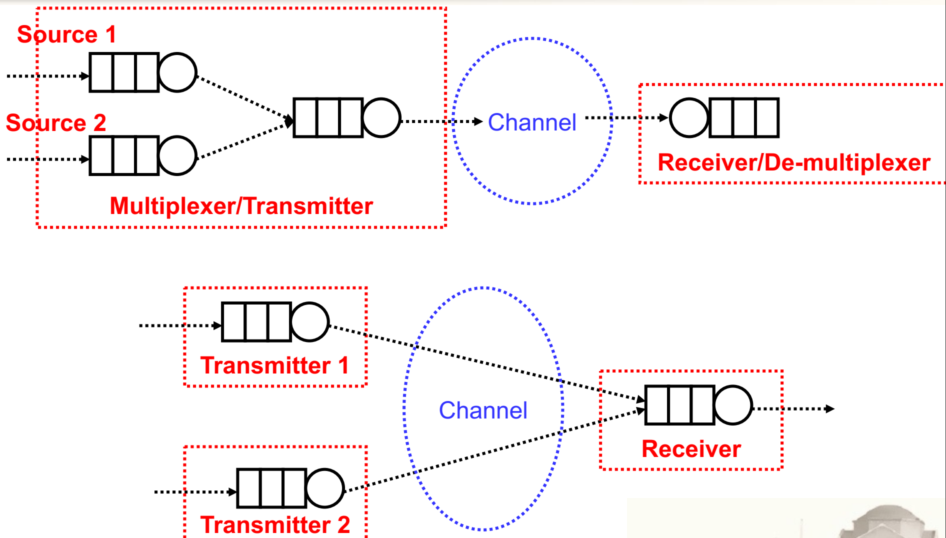
Multi-Access vs. Multiplex



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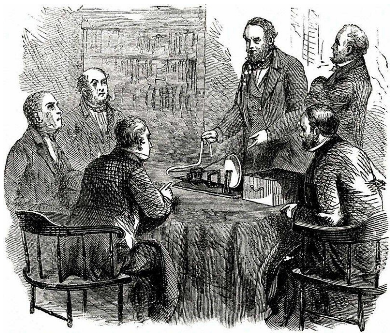
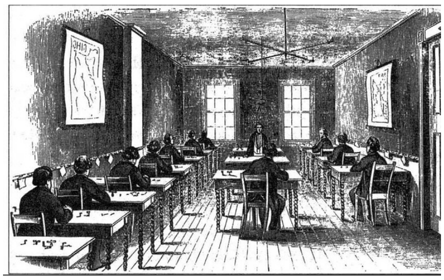
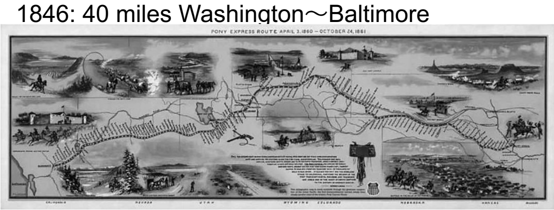
Who drive Duplex?





America, Telegraph Station
(700 miles, 30 station, 200 telegrapher)

1844, Morse Code and Telegraph
1846: 40 miles Washington～Baltimore



1852: 23,000 miles in America
1861, cover east to west

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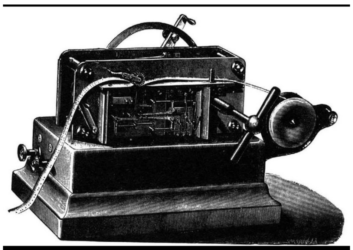
Who drive Duplex?



1858, Whestone, automatic Telegraph machine

- 1872, duplex, Joseph B. Sterns, America
- 1874, Multiplex 6 （in time）, Gene Morris Emil bodo, France

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Contents

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- Resource Divide Modes
- Multi-Access: Fix-allocation, Controlled Access, Random Access
- Multiplex: Time-multiplex (SDH)

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Resource Partition

- Duplex：
  - Time-Division-Duplex (TDD)
  - Frequency-Division-Duplex (FDD)
- Multi-Access
  - Time-Division-Multiple-Access (TDMA)
  - Frequency-Division-Multiple-Access (FDMA)
  - Code-Division-Multiple-Access (CDMA)
  - Space-Division-Multiple-Access (SDMA)
- Multiplex
  - (Time) Synchronous Digital Hierarchy (SDH)
  - (Frequency) Wavelength Division Multiplexing (WDM)

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Channelization

- **Definition and Application**
  - The available bandwidth of a link is shared in time, frequency, or through code, among different stations.
  - It has been used in cellular phone systems.
- **Taxonomy**
  - Frequency-division multiple access (FDMA)
  - Time-division multiple access (TDMA)
  - Code-division multiple access (CDMA)

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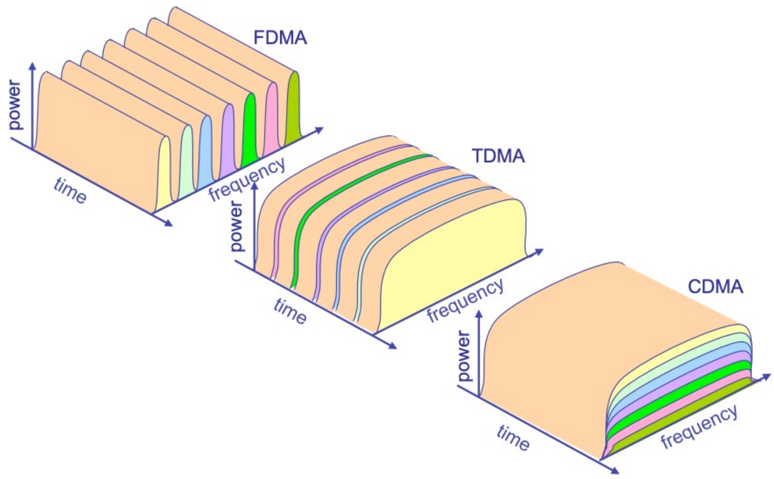
Partitioning Visualization



Courtesy Takashi Inoue

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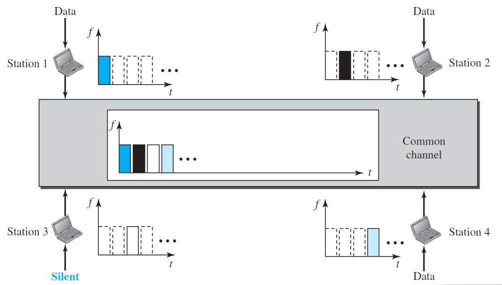
Time-Division Multiple Access

Introduction

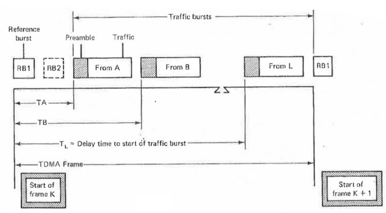
- The stations share the bandwidth of the channel in time
- Each station transmits its data in its assigned time slot
- Guard times are inserted to reduce inference.
- Achieving synchronization between the different stations is the main problem.



Time-Division Multiple Access

TDMA frame organization

- Each frame is divided into reference bursts and traffic bursts
- The position and duration of each traffic burst is assigned to different node according to a protocol established for network operation
- Specific structure of reference bursts ensures the synchronization of all nodes in a network
- Frame efficiency: the ratio of the number of symbols available for carrying traffic to the total number of symbols available in the TDMA frame



Synchronization of TDMA

- Unique Word (UW) Correlation
  - The UW is a common sequence in each frame. At a receiver, the UW is supplied to a UW correlator, where it is correlated with a stored pattern of itself.
  - If we adopt a sequence of length N as UW, the output of the UW correlator is N without any error.
  - In practical application, if the output of correlator is larger than $N-E$, this UW can be detected. E is the error threshold of the UW correlator.

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Synchronization of TDMA

- Unique Word (UW) Correlation
  - Two types of error
    - Miss: the failure of the UW detector to produce a correlation spike that attains or exceeds a preassigned correlation threshold
    - False alarm: the occurrence of a UW just hit from the correlator when the UW is not actually present

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Synchronization of TDMA

Unique Word (UW) Correlation

- Assume the channel is BSC, the bit error probability is p
- The probability of a miss is given by

$$Q = \sum_{i=E+1}^{N} \binom{N}{i} p^i (1-p)^{N-i}$$

- Assuming that all possible receiving sequence are equally likely to occur, the probability of a false alarm is given by

$$F = 2^{-N} \sum_{i=0}^{E} \binom{N}{i}$$

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Synchronization of TDMA

Multiple Unique Words Correlation

- A successful declaration of the UW hit requires that all K UWs achieve their correlation thresholds.
- Assume that the probability of a miss and a false alarm for one UW is Q and F, the probability of miss for K UWs is

$$Q_K = 1 - (1-Q)^K \approx KQ$$

- The probability of false alarm for K UWs is

$$F_K = F^K$$

- If the capacity of the K UWs is used to form only one UW, the false alarm probability will be equal to or less than that given by the above equation.

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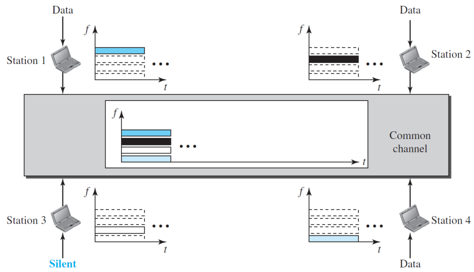
Frequency-Division Multiple Access

⊕ Introduction

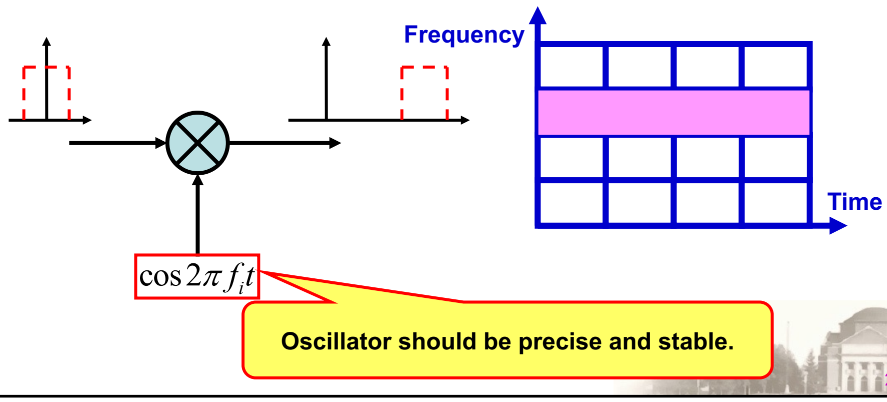
- ☐The available bandwidth is divided into frequency bands
- ☐Each band is reserved for a specific station
- ☐To prevent station interferences, the allocated bands are separated from one another by small guard bands



Frequency-Division Multiple Access

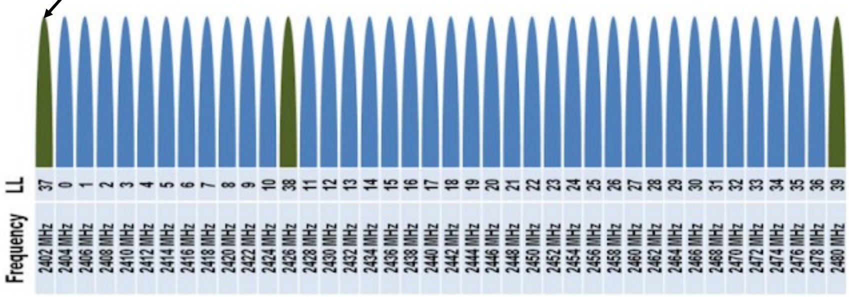
⊕ Frequency division

- ☐Mixer, frequency up/down converter
- ☐Oscillator, upper frequency transformer, wave filter,



Case: Channels in Bluetooth

Preamble code in broadcast channel：10010110



Digital Radio：Tune/Tuning

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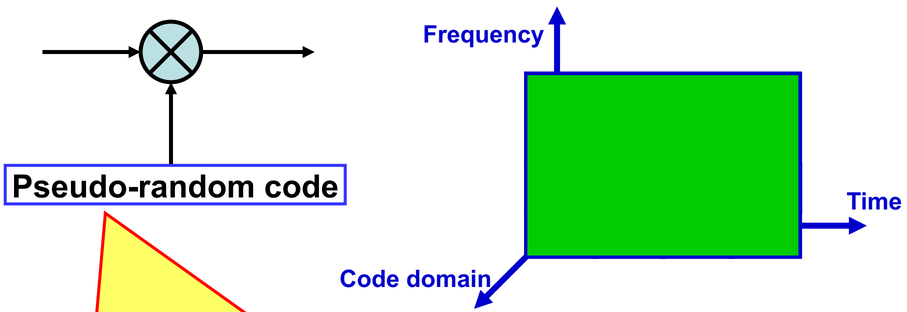
Code Division Multiple Access

Code Division

- Spread spectrum of a signal, using different Pseudo-random codes



Pseudo random code generator vs. frequency mixer
Much higher precise oscillator required

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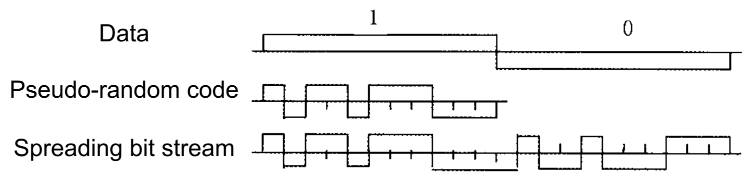
Code Division Multiple Access

Introduction

- Spreading: spreading of the signal is accomplished in the transmitter by means of an independent spreading signal, such that the resulting spread spectrum signal occupies a bandwidth much larger than the bandwidth of the original information-bearing signal.
- Despreading : despreading is achieved by correlating the received signal with a synchronized replica of the spreading signal in the receiver.



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Code Division Multiple Access

Do nothing to physically separate the channels

- All stations transmit at same time in same frequency bands
- One of so-called spread-spectrum techniques

Sender modulates their signal on top of unique code

- Sort of like the way Manchester modulates on top of clock
- The bit rate of resulting signal much lower than entire channel

Receiver applies code filter to extract desired sender

- All other senders seem like noise with respect to signal

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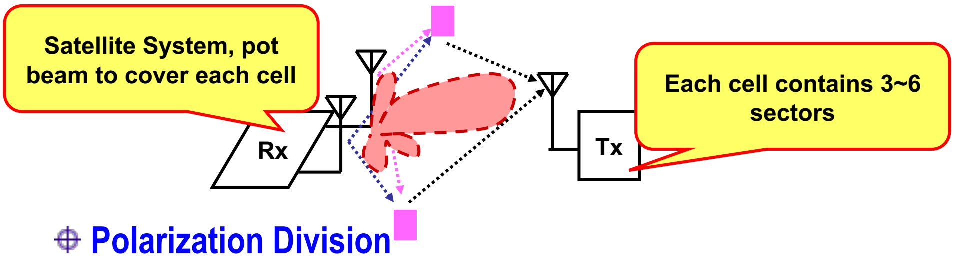
Space/Polarization partition

- **Space Division**
  - Antenna array, directional beam,
  - No ideal beam forming exist.



- **Polarization Division**
  - orthogonal polarization to reuse spectrum resources
  - Most antennas show different gain on different polarization direction (vertical or horizontal)

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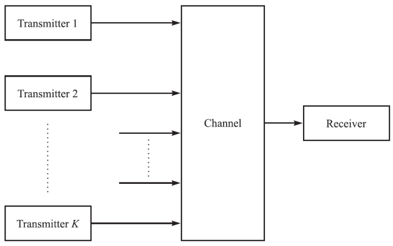
Multiple Access

- **Definition**
  - Multiple users/nodes share a common communication channel to transmit information to a receiver.

- **Taxonomy of Multiple-Access Technologies**
  - Channelization (Channel Partition)
  - Controlled Access
  - Random Access



Capacity of Multiple Access Methods

- **Achievable rates of FDMA and TDMA**
  - Take the following system as an example
    - An ideal AWGN channel of bandwidth W
    - The power spectral density of the additive noise is $N_0/2$
    - K users
    - Each user has an average power $P_i = P, 1 \leq i \leq K$
    - The capacity of a single user is

$$C = W \log_2\left(1 + \frac{P}{WN_0}\right)$$

Capacity of Multiple Access Methods

- **FDMA**
  - Each user is allocated a bandwidth W/K
  - Hence, the capacity of each user is

$$C_K = \frac{W}{K}\log_2\left[1+\frac{P}{(W/K)N_0}\right]$$

  - The total capacity for the K users is

$$KC_K = W\log_2\left(1+\frac{KP}{WN_0}\right)$$

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Capacity of Multiple Access Methods

- **FDMA**
  - The total capacity is equivalent to that of a single user with average power $P_{\text{avg}} = KP$
  - For a fixed bandwidth W, as K increases, the total capacity goes to infinity as the number of users increases. However, the capacity per user decreases.

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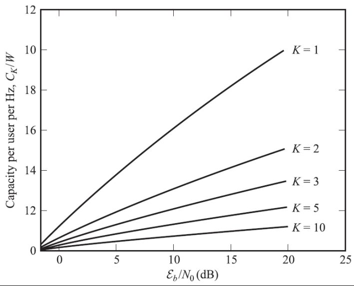
Capacity of Multiple Access Methods

- **Capacity per user normalized by the channel bandwidth W**
  - The average energy of per bit is $E_b = P/C_K$
  - The capacity per user normalized by the channel bandwidth W, as a function of E_b, is given by



$$\frac{C_K}{W} = \frac{1}{K}\log_2\left(1 + K\frac{C_K}{W}\frac{E_b}{N_0}\right)$$

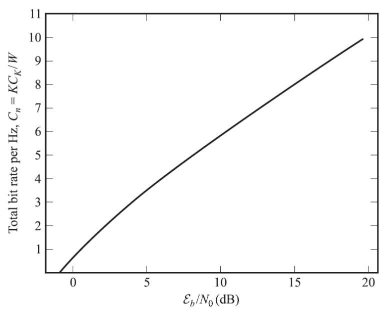
Capacity of Multiple Access Methods

- **Capacity per user normalized by the channel bandwidth W**
  - Defining the normalized total capacity $C_n = KC_K/W$, which is the total bit rate for all K users per unit of bandwidth.
  - Then

$$C_n = \log_2\left(1 + C_n\frac{E_b}{N_0}\right)$$

i.e.

$$\frac{E_b}{N_0} = \frac{2^{C_n} - 1}{C_n}$$



  - C_n increases as E_b/N_0 increases above the minimum value of $\ln 2$

Capacity of Multiple Access Methods

TDMA

- Each user transmits for $1/K$ of the time through the channel of bandwidth W, with average power KP.
- The capacity of per user is

$$C_K = \frac{1}{K} W \log_2 \left(1 + \frac{KP}{WN_0}\right)$$

- This capacity is same with that of an FDMA system
- However, in practical TDMA, it is hard to ensure the transmitters sustain a transmitting power of KP when P is large.

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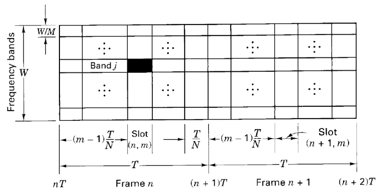
Delay Analysis

Packet Delays in FDMA and TDMA

- Assume an equal apportionment of the total bandwidth W among M nodes. Time axis is partitioned into time frames, each of duration T (means , both TDMA and TDMA will send out 1 unit Packet in T duration.)
- For TDMA, the frames are partitioned into N slot times, each of duration T/M. **Each packet is sent in one time slot *T/M*.**
- FDMA would provide M bands each with a bandwidth of W/M. Each packet is sent over a frame.



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Delay Analysis

Packet Delays in FDMA and TDMA

- The average packet delay is defined as

$$D = w + \tau$$

w: average packet waiting time

τ: packet transmission time

- The packet transmission time for FDMA and TDMA

$$\tau_{FD} = T \qquad \text{(Each packet is sent over a frame } T\text{)}$$

$$\tau_{TD} = T/M \qquad \text{(Each packet is sent over a slot } T/M\text{)}$$

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Delay Analysis

Packet Delays in FDMA and TDMA

- The packet waiting time for FDMA and TDMA
  - The FDMA channel is continuously available and packets are sent as soon as they are generated

$$w_{FD} = 0$$

  - For TDMA, the waiting time of the m-th ($1 \le m \le M$) node is $(m-1)T/M$. Therefore, the average waiting time of TDMA is given by

$$w_{TD} = \frac{1}{M}\sum_{m=1}^{M}(m-1)\frac{T}{M} = \frac{T}{2}\left(1 - \frac{1}{M}\right)$$

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Delay Analysis

Packet Delays in FDMA and TDMA

- The average packet delays of FDMA and TDMA

$$D_{FD} = w_{FD} + \tau_{FD} = T$$

$$D_{TD} = w_{TD} + \tau_{TD} = \frac{T}{2}\left(1 - \frac{1}{M}\right) + \frac{T}{M}$$

- We obtain

$$D_{TD} = D_{FD} - \frac{T}{2}\left(1 - \frac{1}{M}\right)$$

- This result indicates that TDMA has smaller average packet delay than FDMA.

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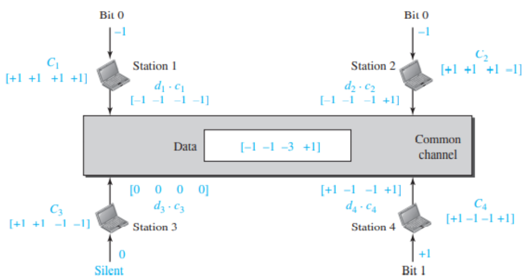
Code Division Multiple Access

Spread-spectrum code (Pseudo-random code)

- Code i is assigned with a unique N-dimensional spread-spectrum code sequence $\mathbf{c}_i$. If the transmit data from node i is d_i (no data then $d_i = 0$), the output is given by

$$\mathbf{y} = \Sigma_{i=1}^{M} d_i \mathbf{c}_i$$



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Code Division Multiple Access

Decoding

- By adopting coherent decoding， the data of node i is given by $u_i = c_i^T y/N$
- If the number of nodes M is not larger than N， orthogonal sequences can be adopted
  - $c_i^T c_j = 0, i \neq j$
  - $c_i^T c_i = N$
- Under this condition, $u_i = d_i$， which means there is no interference between different nodes.
- If $M > N$, we can not assign orthogonal sequences to all nodes. However, non-orthogonal sequences can be designed to reduce the interference between nodes.

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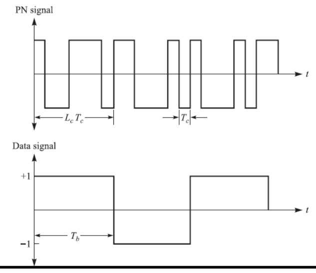
Code Division Multiple Access

Spread factor

- T_b is the bit duration of original data signal
- Adopt N-dimension spread-spectrum code
- T_c is the bit duration of modulated signal, which is called *chip duration*
- $N = T_b/T_c$ is defined as *spread factor*
- Example：BPSK，binary spread-spectrum code



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Code Division Multiple Access

Properties

- The users of a common wireless channel are permitted access to the channel through the assignment of a spreading code to each individual user under spread spectrum modulation.
- CDMA permits the radio resources to be shared among multiple users without requiring the bandwidth allocation of FDMA nor the time synchronization needed in TDMA. FDMA and TDMA can be considered as specifical cases of CDMA.

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Capacity of CDMA

Achievable rate of CDMA

- Taking the following system as an example
  - An ideal AWGN channel of bandwidth W
  - The power spectral density of the additive noise is $N_0/2$
  - K users
  - Each user has an average power $P_i = P, 1 \leq i \leq K$
- The capacity of CDMA depends on the level of cooperation among the K users

  Single-user detection

  Multi-user detection

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Capacity of CDMA

Single-user detection

- The other users' signals appear as interference at the receiver of each user
- The capacity per user for single-user detection is

$$C_K = W \log_2 \left[1 + \frac{P}{WN_0 + (K-1)P} \right]$$

- The capacity per user normalized by bandwidth is

$$\frac{C_K}{W} = \log_2 \left[1 + \frac{C_K}{W} \frac{E_b/N_0}{1 + (K-1)(C_K/W)(E_b/N_0)} \right]$$

- For a large number of users， according to $\ln(1+x) < x$, we obtain

$$\frac{C_K}{W} \leq \frac{C_K}{W} \frac{E_b/N_0}{1 + K(C_K/W)(E_b/N_0)} \log_2 e$$

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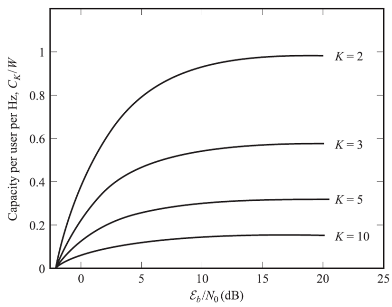
Capacity of CDMA

Single-user detection

- From above analysis, we obtain the normalized total capacity $C_n = KC_kW$ satisfies

$$C_n \leq \log_2 e - \frac{1}{E_b/N_0} < \log_2 e$$



- As we mentioned before, in TDMA/FDMA

$$\frac{E_b}{N_0} = \frac{2^{C_n} - 1}{C_n}$$

- The normalized total capacity increases as E_b/N_0 increases without upper bound

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Capacity of CDMA

Multi-user detection

- Multi-user demodulates and decodes all the users' signals cooperatively, which results in higher capacity.
- Capacity Region
  - Assume the rate of user i is R_i, then the achievable rate of CDMA satisfies

$$R_i < W\log_2\left(1+\frac{P}{WN_0}\right), 1 \le i \le K$$
$$R_i + R_j < W\log_2\left(1+\frac{2P}{WN_0}\right), 1 \le i, j \le K$$
$$\cdots$$
$$\sum_{i=1}^{K} R_i < W\log_2\left(1+\frac{KP}{WN_0}\right)$$

  - For the special case in which $R_i=R, 1\le i\le K$, the above constraints are equivalent to

$$R < \frac{W}{K}\log_2\left(1+\frac{KP}{WN_0}\right)$$

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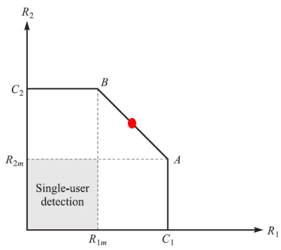
Capacity of CDMA

Example

- The figure shows the capacity region of two-user CDMA

$$C_1 = C_2 = W\log_2\left(1+\frac{P}{WN_0}\right)$$
$$R_{1m} = R_{2m} = W\log_2\left(1+\frac{2P}{WN_0}\right) - C_1$$
$$= W\log_2\left(1+\frac{P}{P+WN_0}\right)$$



- R_{1m} corresponds to the case in which the signal from user 2 is considered as an equivalent additive noise in the detection of the signal of user 1.
- The capacity of single-user detection corresponds to the shaded part of the figure.

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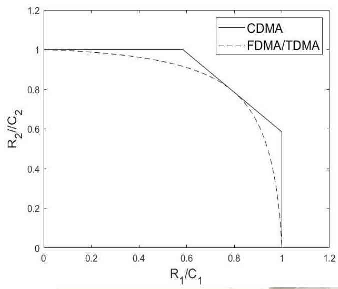
Capacity of CDMA

- **Example**
  - A system adopts FDMA/TDMA
  - Two nodes: Node 1 with bandwidth αW
    Node 2 with bandwidth $(1-\alpha)W$
  - The achievable rates of two nodes are

$$R_1 = \alpha W \log_2\left(1 + \frac{P}{\alpha W N_0}\right)$$

$$R_2 = (1-\alpha) W \log_2\left(1 + \frac{P}{(1-\alpha) W N_0}\right)$$



  - FDMA/TDMA can be seen as a special case of CDMA

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Contents

- Concept： Duplex, Multiplex, Media Access
- Duplex: Half-duplex, Full-duplex
- Resource Divide Modes
- Multi-Access: Fix-allocation, Controlled Access, Random Access
- Multiplex: Time-multiplex (SDH)

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Problem w/Channel partitioning

- Not terribly well suited for random access usage
- Instead, design schemes for more common situations
  - Not all nodes want to send all the time
  - Don’t have a fixed number of nodes
- Potentially higher throughput for transmissions
  - Active nodes get full channel bandwidth

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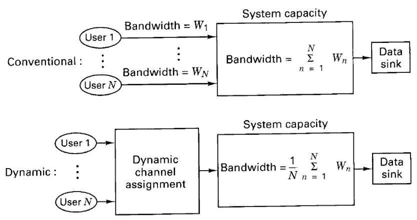
Dynamic Channel Assignment

- Above analysis based on channelization
  - Fixed assignment of resources; total capacity is equal to the sum of the user requirements
- Dynamic channel assignment
  - Give the station access to the channel only when it requests access
  - With bursty traffic and adequate size of buffer, the capacity is equal to the average of the user requirement



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Controlled Access

Definition and application

- The station consult one another to find which station has the right to send. A station cannot send unless it has been authorized by other stations.
- Some of controlled-access protocols are used in LANs.

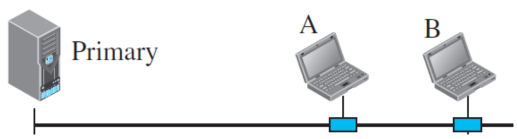
Typical Methods

- Polling
- Token Passing

Polling

Introduction

- System consists of a primary station and many secondary stations
- All data exchanges must be made through the primary device
- The primary device controls the link by select function and poll function.

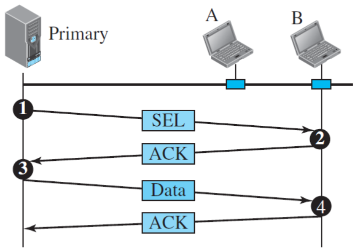


- The drawback is if the primary station fails, the system goes down.

Polling

⌖ Select

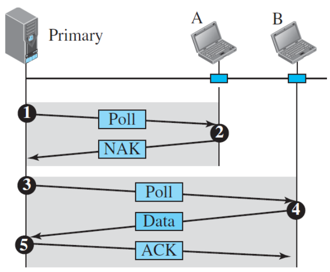
- The primary device alerts the secondary devices to the upcoming transmission and wait for an acknowledgment of the secondary's ready status.



Polling

⌖ Poll

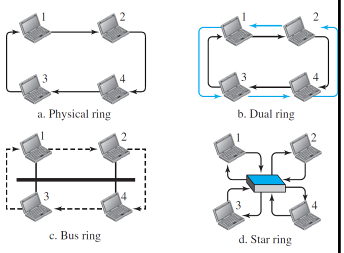
- When the primary is ready to receive data, it must ask (poll) each device in turn if it has something to send.
- If a secondary has nothing to send, then it responds a NAK and the primary polls the next secondary in the same manner.
- If a secondary has a data frame to send, then the primary device reads the frame and returns an ACK.



Token Passing

⊕ Introduction

- The stations in a network are organized in a logical ring. For each station, there is a predecessor and a successor.
- The possession of the token gives the station the right to access the channel and send its data. When a station has some data to send, it waits until it receives the token from its predecessor
- Stations do not have to be physically connected in a ring.



Token Passing

⊕ Token management

- Stations must be limited in the time they can have possession of the token
- The token must be monitored to ensure it has not been lost or destroyed
- Assign priorities to the stations and to the types of data being transmitted

Contents

- Concept： Duplex, Multiplex, Media Access
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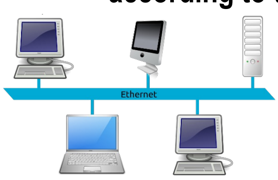
Random Access

- Random Multiple Access
  - Random access are mostly used in LANs and WANs
    - Stations are connected and use a common link, e.g., Ethernet and Wi-Fi
    - No station is superior to another station and none is assigned control over another
    - The station make a decision on whether or not to send according to the state of the medium (idle or busy)



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Random Multiple Access

- Random Multiple Access
  - Properties
    - There is no scheduled time for a station to transmit; Transmission is random among the stations (Random Access)
    - Stations compete with one another to access the medium (Contention)
    - If more than one station tries to send, there is an access conflict and the frames will be either destroyed or modified (Collision)

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Random Multiple Access

- Random Multiple Access
  - To avoid access conflict or to resolve it when it happens, each station follows a procedure that answers the following questions：
    - When can the station access the medium?
    - What can the station do if the medium is busy?
    - How can the station determine the success or failure of the transmission?
    - What can the station do if there is an access conflict?

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ALOHA

- ALOHA——the earliest random access method
  - Developed in early 1970 at the University of Hawaii
    - It can also be used on any shared medium
  - When a station sends data, another station may attempt to do so at the same time. Then data from the two stations collide and become garbled.

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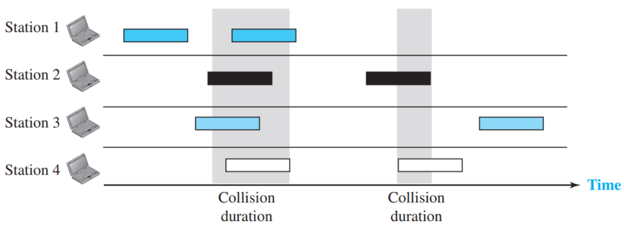
ALOHA

- ALOHA——the earliest random access method
  - Original (pure) ALOHA： each station sends a frame whenever it has a frame to send
  - Schematic diagram of frames in a pure ALOHA network：



  - There are a total of 8 frames; Only two frames survive
  - Even if one bit of a frame coexists on the channel with one bit from another frame, there is a collision and both will be destroyed

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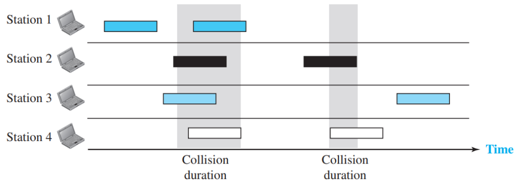
ALOHA Protocol

ACK and retransmission mechanism



- If the receiver receives the frame and decodes it correctly, an acknowledgment (ACK) will be sent to the station.
- If collision happens:

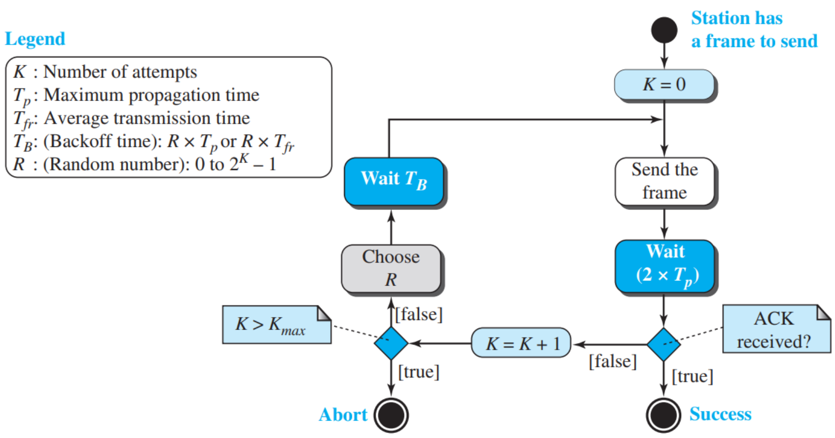
  A) receiver does not receive the frame: ACK does not arrive after a time-out period

  B) receiver detects the error in the frame: receiver sends NAK to the station

Station needs to resend these frames

ALOHA Protocol

Procedure for pure ALOHA protocol



ALOHA Protocol

ACK and retransmission mechanism

- The time-out period is equal to the maximum possible round-trip propagation delay $2T_p$
- If all these stations try to resend their frames after the time-out, the frames will collide again. When the time-out period passes, each station waits a random amount of time before resending its frame, which is called backoff time T_B

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ALOHA Protocol

ACK and retransmission mechanism

- One common formula for T_B is the binary exponential backoff. For each retransmission, a multiplier $R = 0$ to $2^K - 1$ is randomly chosen and multiplied by T_p (maximum propagation time) or T_{fr} (the average time required to send out a frame) to find T_B.
- The range of random numbers increases after each collision. The value of K_{max}, i.e., the maximum value of K, is usually chosen as 15.

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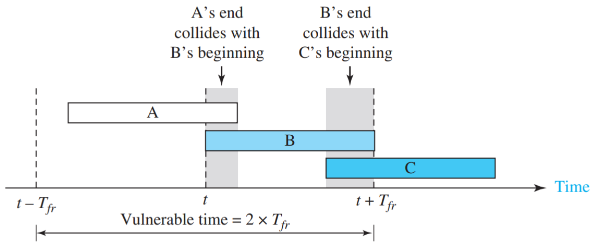
ALOHA Protocol

Vulnerable Time

- Definition：vulnerable time is the length of time in which there is a possibility of collision.



- The vulnerable time for station B is $2T_{fr}$ since there are two possible collision between frames from station B and stations A and C.

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ALOHA Protocol

Throughput

- Assume the average frame arrival rate is λ
  - The system requires an average accepted frames rate equal to λ.
- Because of the presence of collisions， some frames will be unsuccessful or rejected. Define the rejection rate λ_r. The total traffic arrival rate λ_t is given by

$$\lambda_t = \lambda + \lambda_r$$

- Assume the length of each frame is b bits. Then the average amount of successful traffic, which is defined as throughput ρ', is given by

$$\rho' = b\lambda$$

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ALOHA Protocol

Throughput

- Define the total traffic $G' = b\lambda_t$, with the channel capacity designed as R:
  - normalized throughput

$$\rho = \frac{b\lambda}{R} \leq 1$$

  - normalized total traffic

$$G = \frac{b\lambda_t}{R} \leq \infty$$

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ALOHA Protocol

Derivation of the throughput

- Define the transmission time of each frame is

$$T_{\text{fr}} = \frac{b}{R} \text{ seconds/packet}$$

- Now normalized throughput and normalized total traffic can be expressed as

$$\rho = \lambda T_{\text{fr}} \qquad\qquad G = \lambda_t T_{\text{fr}}$$

- The frame arrival statistics for unrelated users of a communication system is often modeled as a Poisson process. The probability of having K new arrival during a time interval of τ seconds is given by

$$P(K) = \frac{(\lambda\tau)^K e^{-\lambda\tau}}{K!}, \quad K \geq 0$$

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ALOHA Protocol

Derivation of the throughput

- A frame is successfully transmitted if there is no arrival during its vulnerable time $2T_{fr}$. Let P_s denote the successful probability, which is given by

$$P_s = P(K=0) = \frac{e^{-2\lambda_t T_{\text{fr}}}}{0!} = e^{-2\lambda_t T_{\text{fr}}}$$

- The probability of a successful frame is $P_s = e^{-2\lambda_t T_{fr}}$. It can also be expressed as

$$P_s = \frac{\lambda}{\lambda_t}$$

- Then we obtain

$$\lambda = \lambda_t e^{-2\lambda_t T_{\text{fr}}}$$

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ALOHA Protocol

Derivation of the throughput

$$\lambda = \lambda_t e^{-2\lambda_t T_{\text{fr}}}$$

- Multiply T_{fr} at both sides, we obtain

$$\rho = Ge^{-2G}$$

  - This equation reveals the relationship between ρ and G
  - The maximum ρ, occurring at a value of $G = 0.5$, is equal to

$$\rho_{\max} = \frac{1}{2e} = 0.184$$

- Intuitive explanation: since the vulnerable time is 2 times the frame transmission time, if a station generates only one frame in its vulnerable time, the frame will reach its destination successfully.

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ALOHA Protocol

- **Example 1:**
  - A pure ALOHA network transmits 200-bit frames on a shared channel of 200 kbps. What is the throughput if the system (all stations together) produces:
    - A) 1000 frames/s
    - B) 500 frames/s
    - C) 250 frames/s
- **Solution:**
  - The frame transmission time is $T_{\text{fr}} = \dfrac{200 \text{ bits}}{200 \text{ kbps}} = 1\text{ms}$
  - A) Normalized total traffic $G = \lambda_t T_{\text{fr}} = 1$

    Normalized throughput $\rho = Ge^{-2G} = 0.135$

    Throughput $\lambda = \rho\lambda_t = 135 \text{ frames/s}$

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ALOHA Protocol

- **Example 1:**
  - A pure ALOHA network transmits 200-bit frames on a shared channel of 200 kbps. What is the throughput if the system (all stations together) produces:
    - A) 1000 frames/s
    - B) 500 frames/s
    - C) 250 frames/s
- **Solution:**
  - B) Normalized total traffic $G = \lambda_t T_{\text{fr}} = 0.5$

    Normalized throughput $\rho = Ge^{-2G} = 0.184$

    Throughput $\lambda = \rho\lambda_t = 184 \text{ frames/s}$

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ALOHA Protocol

Example 1：

- A pure ALOHA network transmits 200-bit frames on a shared channel of 200 kbps. What is the throughput if the system (all stations together) produces：
  - A) 1000 frames/s
  - B) 500 frames/s
  - C) 250 frames/s

Solution：

- C) Normalized total traffic $G = \lambda_t T_{\text{fr}} = 0.25$

  Normalized throughput $\rho = Ge^{-2G} = 0.152$

  Throughput $\lambda = \rho\lambda_t = 152\ \text{frames/s}$

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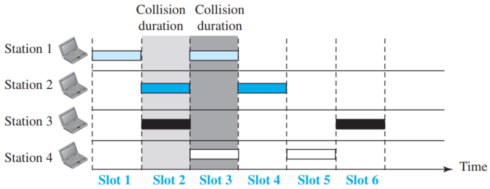
Slotted ALOHA Protocol

Motivation

- The disadvantage of pure ALOHA is the large vulnerable time $2T_{fr}$.
- In slotted ALOHA, we divide the time into slots of T_{fr} seconds and force the station to send only at the beginning of the time slot.



- Because a station is allowed to send only at the beginning of the synchronized time slot, if a station misses this moment, it must wait until the beginning of the next time slot.

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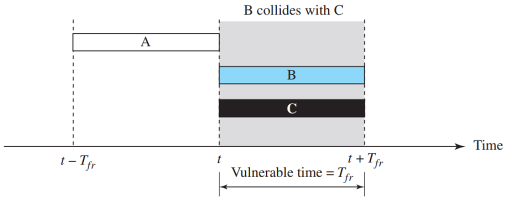
Slotted ALOHA Protocol

Vulnerable time for Slotted ALOHA is T_{fr}



- If two stations try to send at the beginning of the same time slot, collision occurs.
- The vulnerable time is now reduced to one-half
- Slotted ALOHA also adopts ACK and retransmission mechanism
  - After the occurrence of collision, each station waits a random amount of T_{fr}

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Slotted ALOHA Protocol

Derivation： Throughput of Slotted ALOHA Protocol

- Similar to the previous derivation for pure ALOHA, a frame is successfully transmitted if there is no arrival during its vulnerable time T_{fr}. Then, the successful probability is

$$P_s = P(K=0) = \frac{e^{-\lambda_t T_{\mathrm{fr}}}}{0!} = e^{-\lambda_t T_{\mathrm{fr}}}$$

  - Note that in this equation the vulnerable time is T_{fr} instead of $2T_{fr}$
- Further, we obtain $\rho = Ge^{-G}$
  - The maximum ρ, occurring at a value of $G = 1$, is equal to

$$\rho_{\max} = 1/e = 0.37$$

  - This is an improvement of two times the pure ALOHA protocol.

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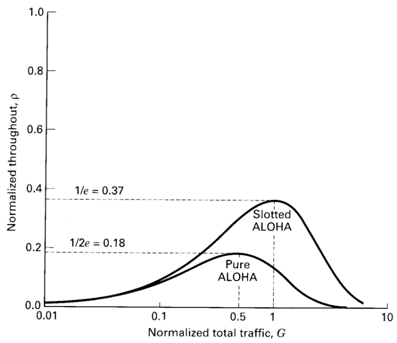
Slotted ALOHA Protocol

- **Performance comparison between slotted and pure ALOHA**
  - **Pure ALOHA** $\rho_{\max} = 1/2e = 0.18$
  - **Slotted ALOHA** $\rho_{\max} = 1/e = 0.37$



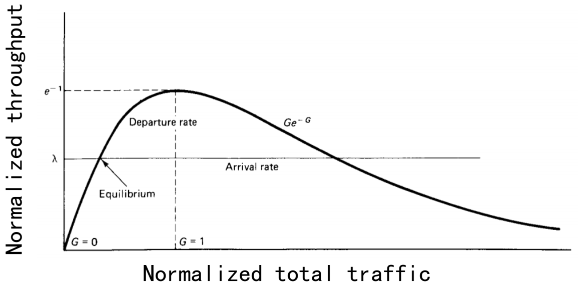
Slotted ALOHA Protocol

- **Example 2:**
  - **Assuming that packet transmissions and retransmission can both be described as a Poisson process, calculate the probability that a data packet transmission in an slotted ALOHA system will experience a collision with one other user. Assume that the total traffic rate $\lambda_t = 10$ packets/s and the packet duration $\tau = 10$ ms.**
- **Solution:**

$$P(K=1) = \left.\frac{(T_{\mathrm{fr}}\lambda_t)^K e^{-T_{\mathrm{fr}}\lambda_t}}{K!}\right|_{K=1}$$
$$= \frac{(0.01 \times 10)^1 e^{-0.01\times 10}}{1!}$$
$$= 0.1e^{-0.1} = 0.09$$

Stability of Slotted ALOHA Protocol

Review the throughput curve of Slotted ALOHA



- For any ρ less than e^{-1}, there are two corresponding values of G
- As the number of backlogged packets changes, the parameter G will change; this leads to a feedback effect, generating further changes in the number of backlogged packets.

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Thanks

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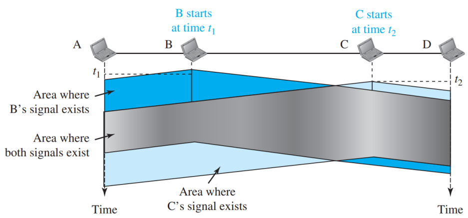
CSMA Protocol

- CSMA (Carrier Sense Multiple Access)
  - Idea: the chance of collision can be reduced if a station senses the medium before trying to use it
    - Listen before talk!

CSMA Protocol

- CSMA (Carrier Sense Multiple Access)
  - CSMA can reduce the possibility of collision, but it can not eliminate it. For example



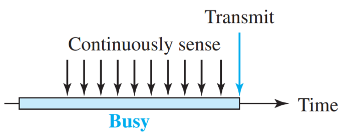
  - Because of propagation delay (although very short), the first bits from station B have not reached station C at time t_2, while station C senses the medium and finds it idle at that time, which results in collision.

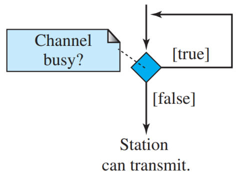
CSMA Protocol

Persistence Methods

- What should a station do if the channel is busy or idle?

1-Persistent

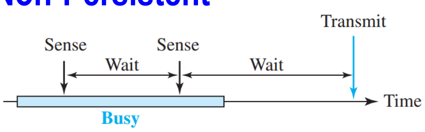


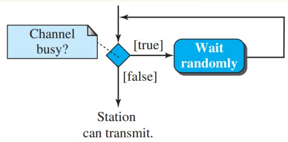


- If the line is busy, the station keeps sensing it
- After the station finds the line idle, it sends its frame immediately (with probability 1)
  - This method has the highest chance of collision because two or more stations may find the line idle and send their immediately.
- Application： Ethernet

CSMA Protocol

Non-Persistent



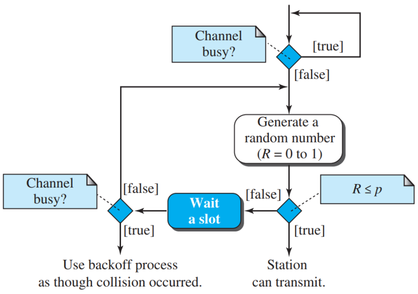


- If a station has a frame to send, then it senses the line:
  - If the line is idle, it sends its frame immediately
  - If the line is not idle, it waits a random amount of time and then senses the line again
- Comment
  - Non-Persistent has less chance of collision than 1-Persistent
  - Non-Persistent also reduces the efficiency of the network because the medium remains idle when there may be stations with frames to send

CSMA Protocol

p-Persistent

- Flow diagram

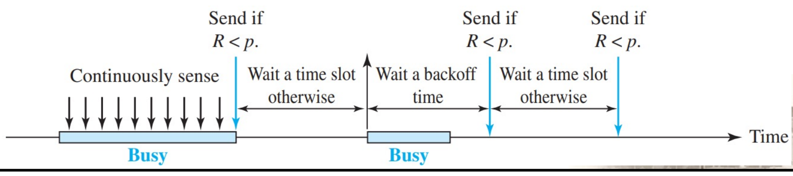


- The p-Persistent approach combines the advantages of the other two strategies. It reduces the chance of collision and improves efficiency.

CSMA Protocol

p-Persistent

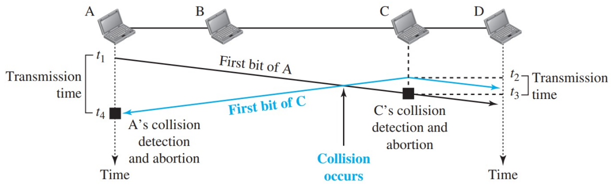
- p-Persistent is used if the channel with a slot duration equal to or greater than the maximum propagation time
- After the station finds the line idle, it follows these steps：
  - Step 1: With probability p, the station sends its frame
  - Step 2: With probability $1 - p$, the station waits for the beginning of the next time slot and checks the line again:
    - If the line is idle, it goes to step 1
    - If the line is busy, it acts as though a collision has occurred and uses the backoff procedure



CSMA/CD Protocol

⊕ CSMA/CD (Collision Detection)

- ☐ Carrier sense multiple access with collision detection
- ☐ A station monitors the medium after it sends a frame to see if the transmission was successful. If there is a collision being detected, the transmission is aborted.



- In the above figure， station A and C are involved in the collision
- Station C detects a collision at time t_3. It immediately aborts transmission
- Station A detects a collision at time t_4. It also immediately aborts transmission

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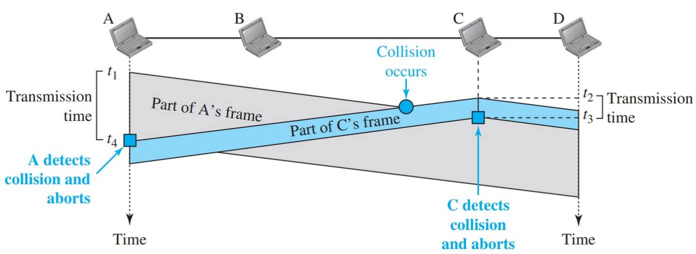
CSMA/CD Protocol

⊕ CSMA/CD (Collision Detection)

- ☐ A station monitors the medium after it sends a frame to see if the transmission was successful. If there is a collision being detected, the transmission is aborted.
- ☐ Collision and abortion in CSMA/CD：



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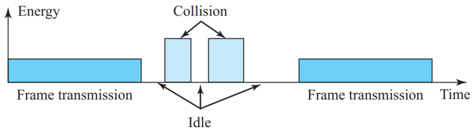
CSMA/CD Protocol

⌖ Energy Level



- ☐ A station that has a frame to send or is sending a frame needs to monitor the energy level to determine if the channel is idle, busy, or in collision mode：
  - At the zero level, the channel is idle
  - At the normal level, a station has successfully captured the channel and is sending its frame
  - At the abnormal level, there is a collision and the level of the energy is twice the normal level
- ☐ According to the energy level， we can determine the state of the channel

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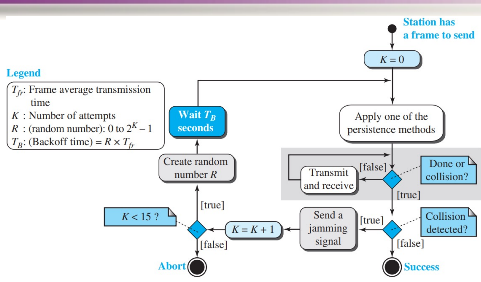
CSMA/CD Protocol



☐ Difference between CSMA/CD and ALOHA:

- Adopt the persistence approach. Sense the channel before starting sending the frame
- During the transmission, station constantly monitors to detect the collision (As shown in the shaded area)
- When the collision is detected， station will send a short jamming signal to make sure that all other stations become aware of the collision

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CSMA/CD Protocol

- **Throughput**
  - The throughput of CSMA/CD is greater than that of pure or slotted ALOHA due to the carrier sense
  - The maximum throughput depends on the value of G and the persistent method
    - For the 1-persistent method, the maximum throughput is around 50% when $G = 1$
    - For the nonpersistent method, the maximum throughput can go up to 90 percent when G is between 3 and 8

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CSMA/CA Protocol

- **Limitation of CSMA/CD：**
  - CSMA/CD works well for wired networks. However, in wireless networks, it is very challenging for a wireless node to listen at the same time as it transmits, since the power of receive signal is orders of magnitude weaker than the transmit signal (its transmission will dwarf any attempt to listen).
- **CSMA/CA：**
  - Carrier sense multiple access with collision avoidance
  - Invented for wireless networks. Collisions are avoided through three strategies：
    - Interframe Space, IFS
    - Contention Window
    - Acknowledgment

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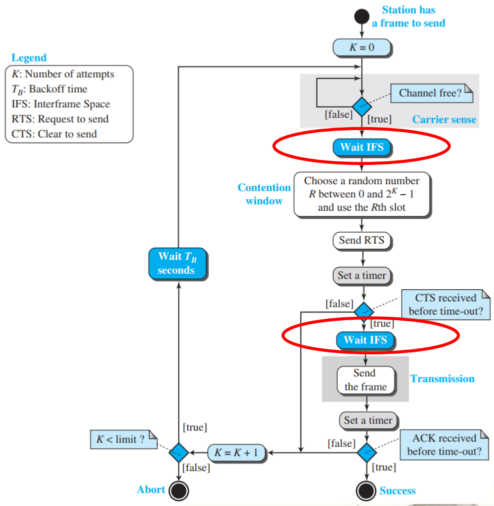
CSMA/CA Protocol

IFS:

- To avoid collisions, the stations waits for a period of time called DCF interface space (DIFS) after the station is found to be idle. If the channel is still idle after waiting a DIFS, the station can send, but it will needs to wait for a time equal to the contention window
- Short interframe space (SIFS) is the amount of time required for a station to process a received frame and to respond with a response frame.



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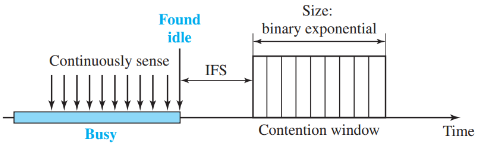
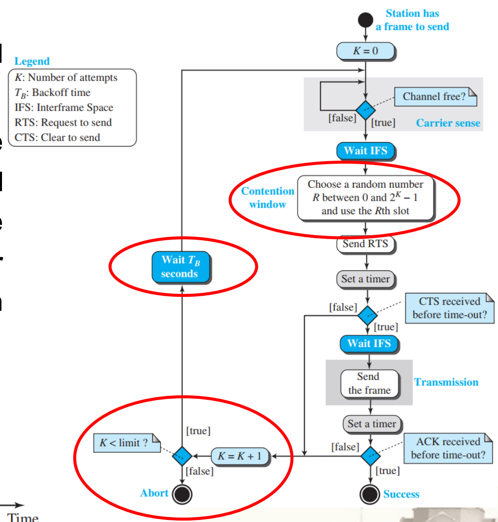
CSMA/CA Protocol

Contention Window (CW):

- Main principle of distributed coordination function (DCF)
- Consists of R time slots, where R is a binary exponential random number in a range $[0, w-1]$. w doubles after each unsuccessful transmission or collision.





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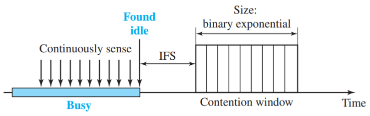
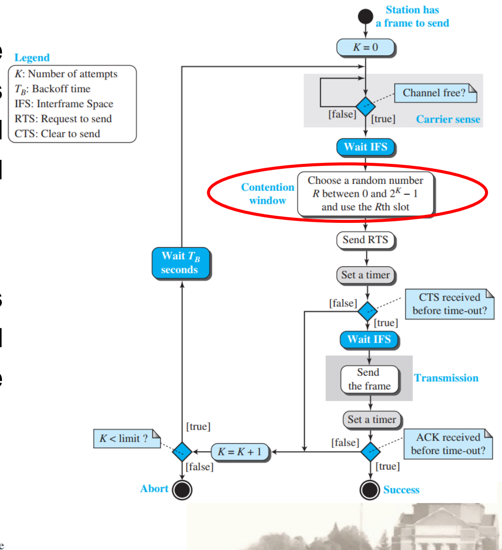
CSMA/CA Protocol

⊕ Contention Window (CW):

- Sense the channel after every time slot: The backoff time counter is decremented as long as the channel is sensed idle. If the channel is still idle after waiting for R time slots, RTS is sent;
- If the channel is busy, the counter is "frozen". And it will be reactivated when the channel is sensed idle again for more than a DIFS.





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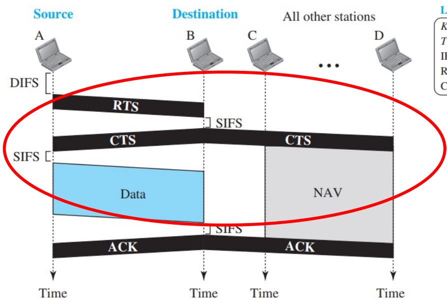
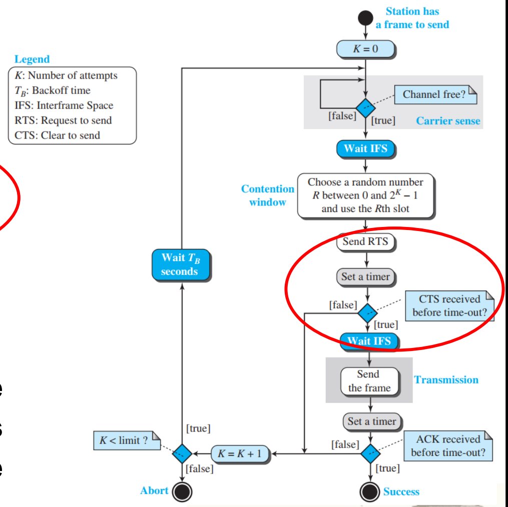
CSMA/CA Protocol

⊕ RTS/CTS Mechanism:



- After the transmitter sending the RTS, it waits for the receiver's CTS, which indicates the receiver is ready to receive data



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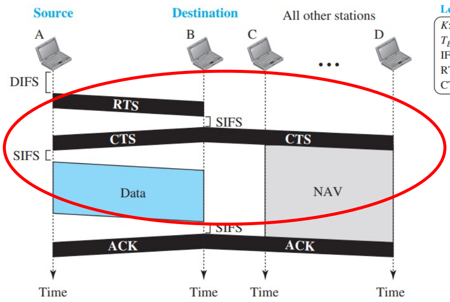
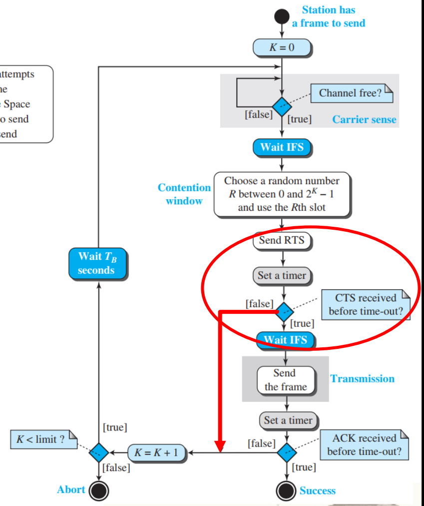
CSMA/CA Protocol

RTS/CTS Mechanism：





- Several stations send RTS together, the collision occurs. The source stations can not receive CTS
- increase the system performance by reducing the duration of a collision when long messages are transmitted

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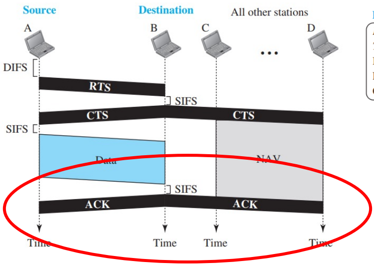
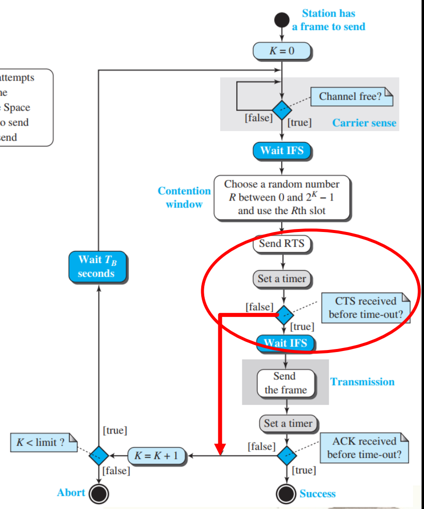
CSMA/CA Protocol

ACK Mechanism：





- **CSMA/CA uses ACK to check for the successful arrival of its data at destination**

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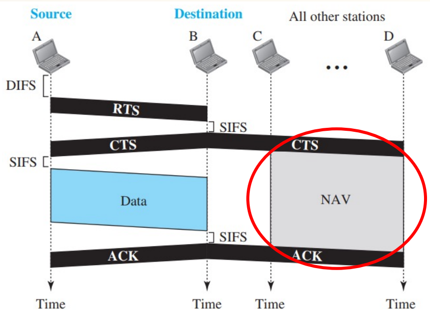
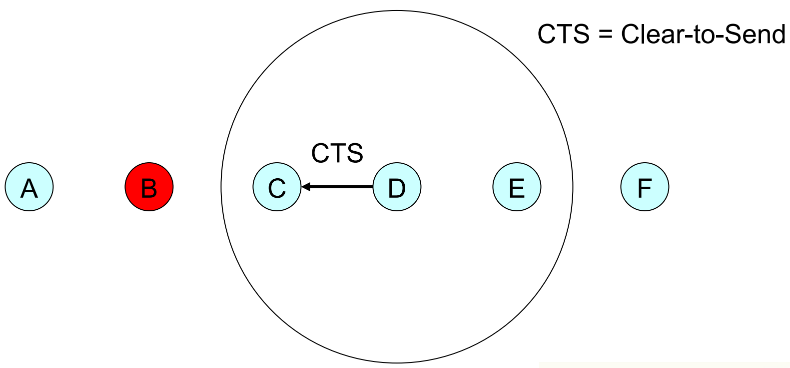
CSMA/CA Protocol

- **Network Allocation Vector, NAV**
  - RTS frame includes the duration of time that it needs to occupy the channel
  - CTS frame also includes the duration of time that it needs to occupy the channel.



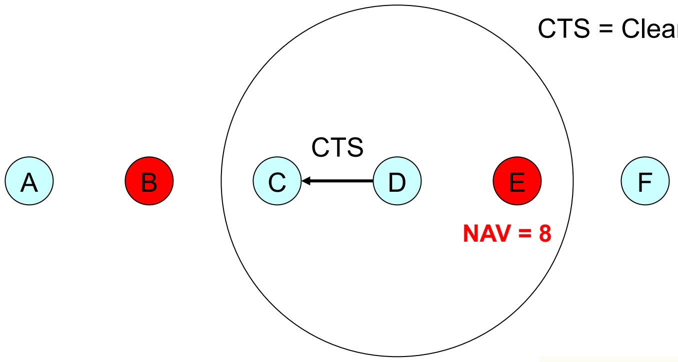
- The stations that are affected by this transmission create a timer called a NAV, which shows how much time must pass before these stations are allowed to check the channel for idleness.
- Each station, before sensing the physical medium to see if it is idle, first checks its NAV to see if it has expired.

IEEE 802.11



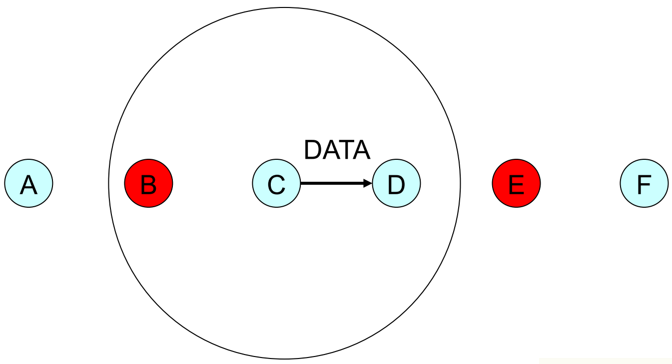
IEEE 802.11

CTS = Clear-to-Send

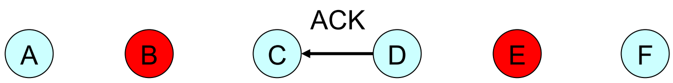


IEEE 802.11

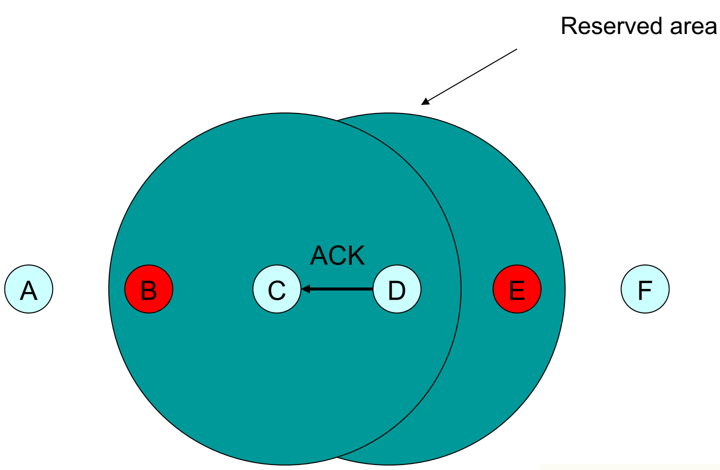
•DATA packet follows CTS. Successful data reception acknowledged using ACK.



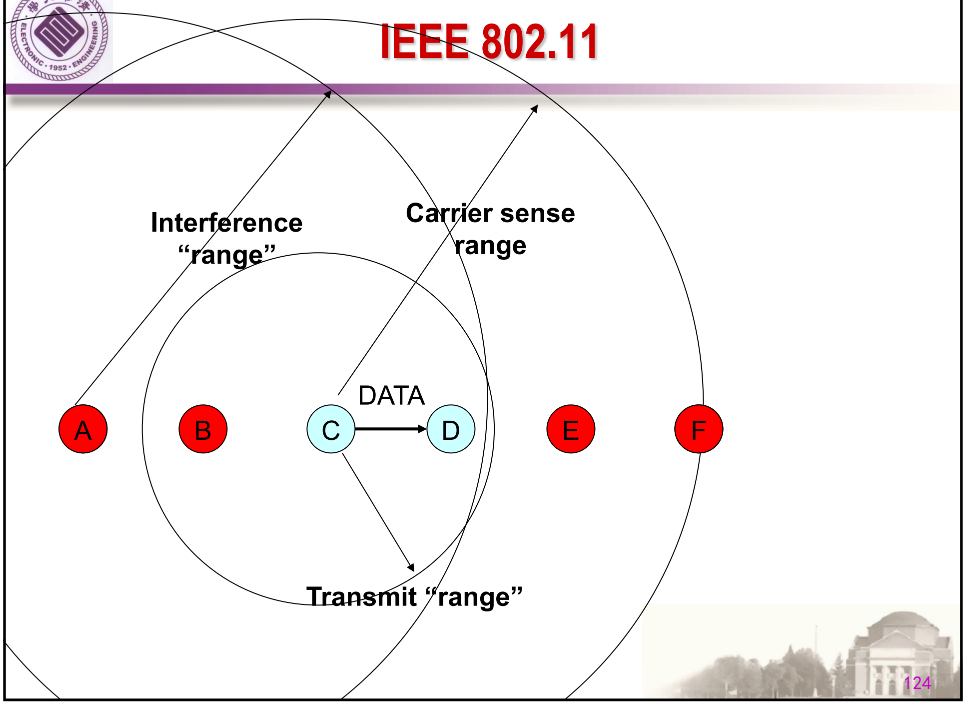
IEEE 802.11



IEEE 802.11



IEEE 802.11



Backoff Interval

- The time spent counting down backoff intervals is a part of MAC overhead
- Important to choose CW appropriately
  - large CW → large overhead
  - small CW → may lead to many collisions (when two nodes count down to 0 simultaneously)

Backoff Interval (Cont.)

- Since the number of nodes attempting to transmit simultaneously may change with time, some mechanism to manage contention is needed
- IEEE 802.11 DCF: contention window CW is chosen dynamically depending on collision occurrence

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Binary Exponential Backoff in DCF

- When a node fails to receive CTS in response to its RTS, it increases the contention window
  - CW is doubled (up to an upper bound)
  - More collisions ➔ longer waiting time to reduce collision
- When a node successfully completes a data transfer, it restores CW to CW_{min}

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802.11 Overhead

- Channel contention resolved using backoff
  - Nodes choose random backoff interval from [0, CW]
  - Count down for this interval before transmission
- Backoff and (optional) RTS/CTS handshake before transmission of data packet
- 802.11 has large room for improvement

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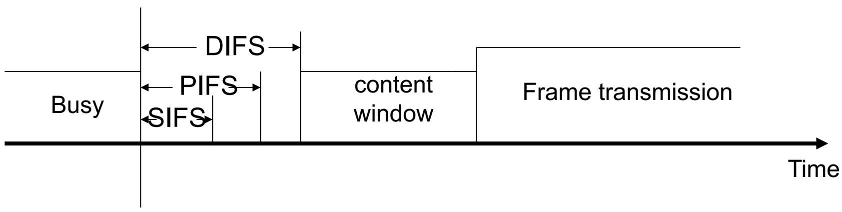
802.11 Frame Priorities



- Short interframe space (SIFS)
  - For highest priority frames (e.g., RTS/CTS, ACK)
- PCF interframe space (PIFS)
  - Used by PCF during contention free operation
- DCF interframe space (DIFS)
  - Minimum medium idle time for contention-based services

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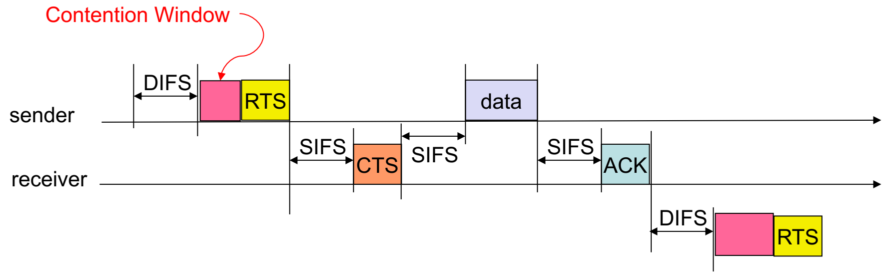
11b/11g Throughput



$$Throughput = \frac{8 \times L_{data}}{T_{DIFS} + 3T_{SIFS} + T_{cw} + \frac{RTS + CTS + DATA + ACK}{R}}$$

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11b/11g Throughput

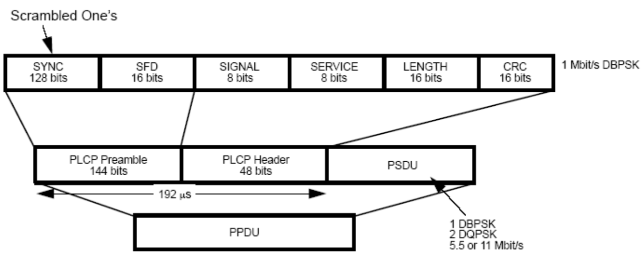

Figure 127—Long PLCP PPDU format

$$Throughput = \frac{8 \times L_{data}}{T_{DIFS} + 3T_{SIFS} + T_{cw} + T_{PHY} + \frac{L_{RTS} + L_{CTS} + L_{DATA} + L_{ACK}}{R}}$$

$$= \frac{8 \times L_{data}}{T_{DIFS} + 3T_{SIFS} + T_{cw} + T_{PHY} + \frac{8 \times 82 + L_{data}}{R}}$$

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802.11 MAC frame

Octets: 2	2	6	6	6	2	6	0 - 2312	4
Frame Control	Duration/ ID	Address 1	Address 2	Address 3	Sequence Control	Address 4	Frame Body	FCS

MAC Header

Table 4 —Address field contents

To DS	From DS	Address 1	Address 2	Address 3	Address 4
0	0	DA	SA	BSSID	N/A
0	1	DA	BSSID	SA	N/A
1	0	BSSID	SA	DA	N/A
1	1	RA	TA	DA	SA

- MAC header + FCS = 34 Bytes
- DATA = 0~2312 Bytes
- 4 Address: BSSID, Destination, Source, Receiver, Transmitter

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MAC frames

Octets: 2	2	6	6	4
Frame Control	Duration	RA	TA	FCS

MAC Header

Figure 16—RTS frame

Octets: 2	2	6	4
Frame Control	Duration	RA	FCS

MAC Header

Figure 17—CTS frame

Octets: 2	2	6	4
Frame Control	Duration	RA	FCS

MAC Header

Figure 18—ACK frame

- Duration (ms)=CTS+ACK+SIFS+DATA
- RTS：20Bytes
- CTS：14Bytes
- ACK：14Bytes
- DATA：34＋(0～2312) Bytes

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11b/11g Throughput

$$Throughput = \frac{8 \times L_{data}}{T_{DIFS} + 3T_{SIFS} + T_{cw} + T_{PHY} + \frac{8 \times 82 + L_{data}}{R}}$$

$$= \frac{8 \times L_{data}}{T_{DIFS} + 3T_{SIFS} + \frac{CW_{MAX}}{2}\sigma + T_{PHY} + \frac{8 \times 82 + L_{data}}{R}}$$

L-data (Byte)	100	200	500	800	1000	1500
Throughput (Mbps)	0.710	1.409	3.441	5.379	6.623	9.574

$DIFS: 50us,\ SIFS: 10us,\ PHY: 192us,\ CW: 31,\ \sigma: 20us,\ R: 11Mbps$

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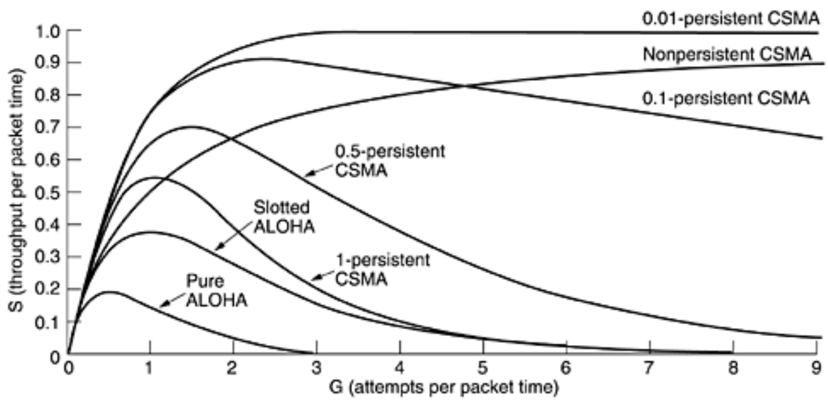
CSMA variants

- 1-persistent CSMA: transmit once no carrier is detected
- *P*-persistent
- Non-persistent CSMA: when carrier is detected, wait a random time before a retry



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Thanks